A Rate-Distortion View of Uncertainty Quantification

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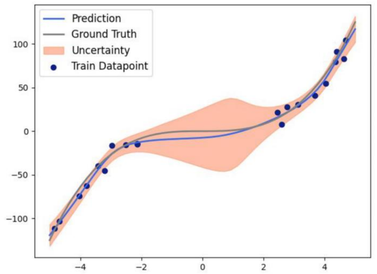
Code and paper!

https://github.com/ifiaposto/Distance_Aware_Bottleneck

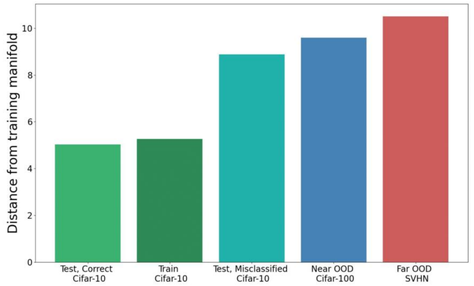
Problem: Efficient Uncertainty Quantification



Goal: reliable & lightweight models that "know what they know."

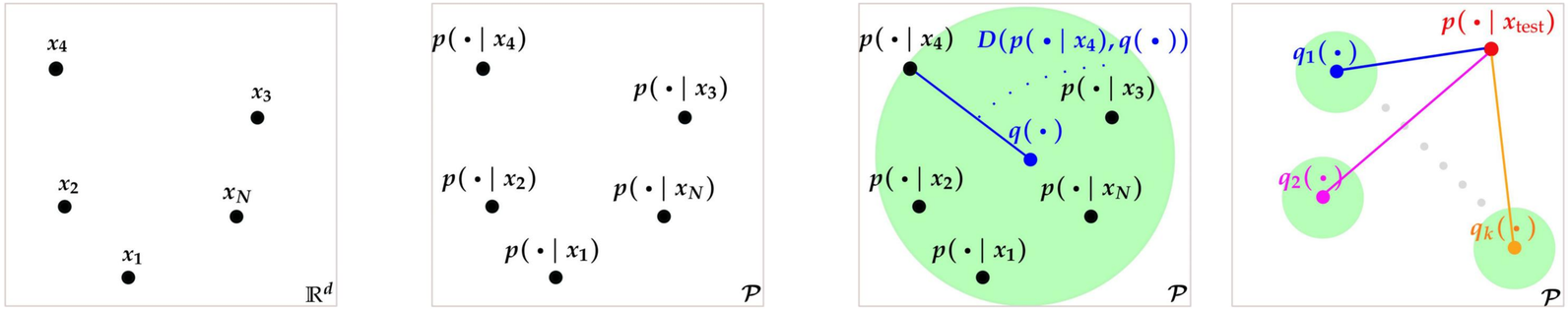
- **Principled**: How much evidence?
- **Efficient**: Single Model & Sampling-Free
- **Post-hoc**: Works on pre-trained models

Key Idea: Distance-Awareness



How far is an input example from the training dataset?

Our Method: Distance-Aware Bottleneck (DAB)

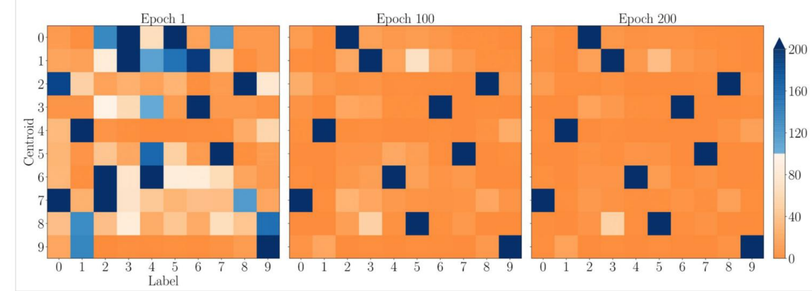


1> Treat the training dataset as a source generating training examples.

2> Map each datapoint to a *distribution* through a stochastic encoder.

3> The codebook contains the centroids of the encoders in terms of a *statistical distance*. Uncertainty is quantified by the expected distance from the codebook: $\text{uncertainty}(x_{\text{test}}) = \mathbb{E}[D(p(z \mid x_{\text{test}}; \theta), q_K(z; \phi))]$

Codebook Visualization



Number of CIFAR-10 test datapoints assigned to each centroid during training. Each centroid progressively attracts datapoints of the same class.

DAB's performance on ImageNet 1-K

Method	Uncertainty Description	Calibration AUROC ↑	OOD AUROC ImageNet-O ↑	Accuracy ↑	# Trainable Parameters
Deep Ensemble of 5	Gibbs softmax entropy	0.861 ± 0.0004	0.642 ± 0.001	$\mathbf{78.4 \pm 0.06\%}$	117,672,960
DAB with fine-tuned ResNet-50 (ours)	Statistical distance (KL)	$\mathbf{0.868 \pm 0.0008}$	$\mathbf{0.743 \pm 0.004}$	$76.1 \pm 0.02\%$	36,612,328
DAB with pre-trained ResNet-50 (ours)	Statistical distance (KL)	0.866 ± 0.0003	0.732 ± 0.004	$74.71 \pm 0.09\%$	**13,077,736**

DAB outperforms ensembles at predicting misclassifications!

DAB can better distinguish out-of-distribution images.

Many fewer parameters!

References

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